

Reinforcement Learning-Based Distributed Secondary Frequency Control and Active Power Sharing in Islanded Microgrids With Bandwidth-Conscious Memory-Event-Triggered Mechanism

Shen Yan[✉], Zehao Dou, and Zhou Gu[✉], *Senior Member, IEEE*

Abstract—This paper studies the reinforcement learning-based distributed secondary frequency control and active power allocation of islanded microgrids under event-triggered mechanism. First, a novel bandwidth-conscious memory-event-triggered mechanism is proposed to reduce communication and computation burdens, which introduces the mean of memory signals to smooth the measured outputs perturbed by disturbances and noises. Meanwhile, a dynamic triggering threshold depending on real-time bandwidth status is constructed to adaptively adjust the data transmission rate according to the changes in bandwidth status and system responses. Second, a new reinforcement learning-based distributed secondary controller using Q-learning algorithm is presented to dynamically regulate the frequency controller gains in response to complex system environments, which helps in achieving better frequency restoration performance. Third, the system stability is analyzed by some linear matrix inequality conditions. Then, simulation outcomes based on load variation and plug-and-play test illustrate that our strategy achieves satisfactory performance in frequency restoration and active power sharing. In addition, some comparison results show the merits of the constructed event-triggered scheme and Q-learning-based distributed secondary frequency controller.

Note to Practitioners—This article is concerned with the reinforcement learning-based distributed secondary frequency control and active power allocation of islanded microgrids, addressing challenges induced by limited communication resources, dynamic bandwidth status and complex operation environment. The innovative bandwidth-conscious memory-event-triggered scheme increases the usage efficiency of communication network, and the reinforcement learning-based distributed secondary frequency controller optimizes the frequency restoration performance for islanded microgrids under load changes and random

plug and unplug behaviors. The developed strategy is able to handle these realistic problems in practical systems. This strategy can provide technical support for electrical engineers to enhance the stability performance of islanded microgrids.

Index Terms—Islanded microgrids, bandwidth-conscious memory-event-triggered mechanism, Q-learning-based control, frequency restoration, active power sharing.

I. INTRODUCTION

MICROGRIDS are small-scale power generation and distribution systems that integrate distributed generation (DG) such as solar, wind, and tidal energy, along with energy storage devices, and loads [1], [2], [3]. When operating in grid-connected mode, microgrids can adjust and optimize power generation according to the requirements of the main grid. In islanded mode, microgrids can operate independently during grid failures or emergencies, providing continuous power supply to users [4], [5]. When a microgrid operates in islanded mode, appropriate control strategies are needed to maintain stable operation and power quality [3], [6].

Currently, hierarchical control is widely employed in microgrids to regulate their operation [7], [8], the architecture of which is divided into three levels: primary control, secondary control and tertiary control [9], [10]. Primary control mainly includes voltage and current dual closed-loop controllers combined with droop control, which leads to voltage/frequency deviations. Secondary control is designed to compensate for the frequency and voltage deviations generated by the primary control, while tertiary control focuses on energy management and economic dispatch. In recent years, to ensure the cooperation of multi nodes, scholars have proposed various distributed secondary control schemes based on information exchange [8], [12], [13], [14], [15]. For instance, [14] investigates the distributed secondary voltage control issue of islanded microgrids subject to time-varying communication delays by using proportional-integral observer. However, the implementation of such distributed control schemes heavily relies on real-time communication among DGs.

The communication networks of microgrids typically have limited channel capacity and bandwidth. In this case, the traditional periodic sampling-based control strategy may transmit

Received 2 July 2025; revised 26 September 2025, 9 December 2025, and 19 January 2026; accepted 23 February 2026. Date of publication 27 February 2026; date of current version 6 March 2026. This article was recommended for publication by Associate Editor G. Ferro and Editor Q. Zhao upon evaluation of the reviewers' comments. This work was supported in part by the National Natural Science Foundation of China under Grant 62573234 and Grant 62273183 and in part by Jiangsu Province Youth Science and Technology Talent Support Project under Grant JSTJ-2025-061. (Corresponding authors: Shen Yan; Zhou Gu.)

Shen Yan and Zehao Dou are with the College of Mechanical and Electronic Engineering, Nanjing Forestry University, Nanjing 210037, China (e-mail: yanshenzdh@gmail.com).

Zhou Gu is with the School of Electrical Engineering, Anhui Polytechnic University, Wuhu 241000, China (e-mail: gzh1808@163.com).

Digital Object Identifier 10.1109/TASE.2026.3668833

1558-3783 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

Authorized licensed use limited to: Anhui Polytechnic University. Downloaded on March 30, 2026 at 00:23:00 UTC from IEEE Xplore. Restrictions apply.

some unnecessary signals, which leads to network congestion and the degradation of network quality of service (QoS) [16], [17]. Therefore, an event-triggered mechanism (ETM) has been utilized in microgrids to reduce the communication frequency [18], [19], [20], [21], [22]. To name a few, some distributed secondary control mechanisms for islanded microgrids based on normal ETM [20] and dynamic ETM [21] are proposed to realize frequency restoration and active power allocation. Note that the triggering conditions in [20] and [21] are considered with static triggering thresholds, which cannot flexibly adjust the data transmission rate in response to the changes of system operating conditions. Recently, some adaptive ETMs with dynamic triggering thresholds have been considered in [23], [24], and [25] to overcome this drawback. However, these dynamic thresholds are designed solely based on system control performance, while neglecting the impact of the time-varying bandwidth status on both triggering performance and control performance. It is investigated in [26] that the triggering thresholds should be related to the real-time bandwidth status which can be indexed by QoS. Specifically, when QoS is ideal, triggering threshold can be decreased to generate more signals to improve control performance. On the other hand, it should be increased to mitigate the degradation of QoS. It is worth noting that, most of the above ETMs rely on current sampled information, which may be perturbed by load variations and noise, resulting in redundant triggering events that waste limited bandwidth resources. Therefore, how to design an appropriate ETM has the ability to adapt to the changes of system evolution and bandwidth status, as well as to reduce redundant triggering events is the first motivation of this work.

On the other hand, microgrids are often deployed in diverse and frequently changing environments such as load variations, random connected/disconnected behaviors, and aperiodic data communications. However, most of existing distributed secondary control methods for microgrids dependent on fixed control gains [14], [19], [20], [21], which lack the adaptability required for such complex environments. The static nature of these methods may lead to degraded control performance or even system instability. To address this limitation, reinforcement learning (RL), a machine learning method where agents learn optimal policies through trial-and-error interactions with environments, offers a promising solution [27], [28]. Unlike traditional supervised learning dependent on labeled datasets, RL leverages feedback mechanisms to dynamically adjust strategies, making it particularly suited for adaptive optimization in microgrid control. To our best knowledge, few results about RL-based distributed secondary event-triggered control for islanded microgrids have been reported to improve control performance and save more communication resources, which is the second motivation.

Inspired by the aforementioned contents, this paper proposes an RL-based distributed secondary event-triggered control strategy for islanded microgrids. The main differences and contributions are as below.

- 1) First, an innovative bandwidth-conscious memory event-triggered mechanism (BCMETM) is proposed to improve the communication network utilization

efficiency. Compared to the existing ETM [26] that rely on instantaneous measurements, our BCMETM utilizes historical system states, effectively reducing redundant triggering events caused by data perturbations. Meanwhile, the triggering threshold in [26] depends on two auxiliary and recursively scalar functions, while our scheme only needs one adaptive triggering threshold function, which decreases the computational burden and enhances deployment convenience. Different from the adaptive triggering thresholds in [24] and [25] independent on network status, our BCMETM adaptively adjusts its triggering threshold based on real-time bandwidth availability and system dynamics.

- 2) Second, an RL-based optimized distributed secondary event-triggered control scheme is developed. Unlike many existing secondary frequency control scheme with fixed controller gains in [14], [19], [20], and [21], our RL-based optimized control scheme employs a Q-learning algorithm to dynamically adjust the controller gains adapting to complex operation conditions. This adaptability enhances both transient response performance and steady-state stability, even under stochastic renewable generation and load fluctuations. Unlike the existing RL-based control method [28], this paper provides rigorous stability guarantees through Lyapunov stability analysis and LMI conditions, ensuring the convergence of both event-triggered control and RL gain adjustment.

Notation: In this work, $\langle Q | x(t) \rangle$ represents $x^T(t)Qx(t)$, where $x^T(t)$ denotes the transpose of $x(t)$. $[x_i(t)]_{i=a,\dots,b} = [x_a^T(t) \dots x_i^T(t) \dots x_b^T(t)]^T$.

II. MICROGRID CONTROL SYSTEM DESCRIPTION

A. Graph Theory

The microgrid system can be represented as a undirected graph $\mathbb{G} = (\mathbb{V}, \mathbb{E})$. Assuming the microgrid system has n DGs, each DG can be regarded as a node. $\mathbb{V}$ is the set of all distributed generation nodes, labeled as $\mathbb{V} = \{1, 2, \dots, n\}$. $\mathbb{E} \in \mathbb{V} \times \mathbb{V}$ is the set of all directed edges that connect the nodes in the graph. The adjacency matrix $\mathbb{A}$ is used to represent the connection relationships between nodes. $\mathbb{A} = [a_{ij}]$ assigns $a_{ij} = 0$ to non-adjacent nodes and $a_{ij} = 1$ for connected nodes i and j . $N_i = \{v_j \in \mathbb{V} | a_{ij} = 1\}$ stands for the set of neighbors of the DG i . The degree matrix can be represented as $\mathbb{D} = \{d_1, d_2, \dots, d_n\}$. Laplacian matrix $\mathbb{L}$ is obtained via $\mathbb{L} = \mathbb{D} - \mathbb{A}$. $\mathbb{B} = \text{diag}\{b_1, b_2, \dots, b_i, \dots\}$, $b_i = 1$ and $b_i = 0$ mean DG i is connected and disconnected with the reference signal, respectively. Assume that at least one DG is able to obtain the reference signal.

B. Control Structure

Fig. 1 shows the co-design framework of the BCMETM and secondary controller for the i -th DG in an islanded microgrid. In order to guarantee the stability of frequency and achieve power sharing, the frequency droop control combined with the secondary control z_i of DG i is designed as

$$\omega_i = \omega' - m_i (P_i - P_i') + z_i, \quad (1)$$

utilized Simpson's second rule in this work is more accurate, which helps decrease the approximation error and design conservatism at the cost of high computation consumption.

D. Distributed Secondary Event-Triggered Control Systems

Under the proposed BCMETM, distributed secondary controllers for frequency regulation and active power sharing are proposed as follows:

$$\dot{z}_i^\omega = k_\omega \sum_{j \in N_i} a_{ij} \left(\bar{\omega}_i(t_{\omega,r}^i h) - \bar{\omega}_j(t_{\omega,r}^j h) \right) + b_i k_\omega (\bar{\omega}_i(t_{\omega,r}^i h) - \omega'), \quad (8)$$

$$\dot{z}_i^P = k_P \sum_{j \in N_i} a_{ij} \left(m_i \bar{P}_i(t_{P,r}^i h) - m_j \bar{P}_j(t_{P,r}^j h) \right), \quad (9)$$

for $t \in [t_{\theta,r}^i h, t_{\theta,r+1}^i h)$, where $k_\omega > 0$ and $k_P > 0$ are the controller gains.

By introducing $\bar{\omega}(t)$ into (2) and defining $\tilde{\omega}_i \triangleq \bar{\omega}_i - \omega'$, $\dot{z}_i^\omega$ is transformed as

$$\dot{\tilde{\omega}}_i = \dot{z}_i - m_i \dot{P}_i = \dot{z}_i^\omega. \quad (10)$$

In terms of (2)–(10), for $t \in [d_s, d_{s+1})$, the closed-loop frequency control and active power sharing control systems are obtained as

$$\dot{\tilde{\omega}}_i(t) = k_\omega \sum_{j \in N_i} a_{ij} \left(\tilde{\omega}_i(t_{\omega,r}^i h) - \tilde{\omega}_j(t_{\omega,r}^j h) \right) + b_i k_\omega \tilde{\omega}_i(t_{\omega,r}^i h), \quad (11)$$

$$m_i \dot{\tilde{P}}_i(t) = k_P \sum_{j \in N_i} a_{ij} \left(m_i \tilde{P}_i(t_{P,r}^i h) - m_j \tilde{P}_j(t_{P,r}^j h) \right). \quad (12)$$

According to Simpson's second rule, $\bar{\theta}_i(d_{s-1})$ can be approximated by

$$\bar{\theta}_i(d_{s-1}) \approx \frac{1}{8} \left[\theta_i(t - \lambda_0(t)) + 3\theta_i(t - \lambda_1(t)) + 3\theta_i(t - \lambda_2(t)) + \theta_i(t - \lambda_3(t)) \right], \quad (13)$$

where $\lambda_l(t) = d(t) + l\tau/3$, $l = 0, 1, 2, 3$, and $d(t) \in [h, 2h]$ is defined as

$$d(t) = \begin{cases} h, & t \in [0, h) \\ t - d_0, & t \in [h, 2h) \\ \dots & \dots \\ t - d_{s-1}, & t \in [d_s, d_{s+1}) \\ \dots & \dots \end{cases} \quad (14)$$

With the help of (13), a unified closed-loop event-triggered control model of frequency restoration (11) and active power sharing (12) is formulated as

$$\begin{aligned} \dot{\tilde{\theta}}(t) = & K_\theta \mathbb{U}_\theta \left[\frac{1}{8} \tilde{\theta}(t - \lambda_0(t)) + \frac{3}{8} \tilde{\theta}(t - \lambda_1(t)) \right. \\ & \left. + \frac{3}{8} \tilde{\theta}(t - \lambda_2(t)) + \frac{1}{8} \tilde{\theta}(t - \lambda_3(t)) - e_\theta(d_s) \right], \end{aligned} \quad (15)$$

where $\mathbb{U}_\theta = \mathbb{L} + \mathbb{B}$ for $\theta_i = \omega_i$, $\mathbb{U}_\theta = \mathbb{L}$ for $\theta_i = m_i P_i$, $\tilde{\theta}(t) = [\tilde{\theta}_i(t)]_{i=1,\dots,n}$, $e_\theta(t) = [e_{\theta_i}(t)]_{i=1,\dots,n}$, $K_\theta = \text{diag}\{k_\theta, \dots, k_\theta\}$.

Lemma 1: [33] For given positive scalars $w, o, \varrho \in (0, 1)$, a positive matrix $\mathbb{W} \in \mathbb{R}^{w \times w} > 0$, matrices $\mathbb{J}_1, \mathbb{J}_2 \in \mathbb{R}^{w \times o}$, and any vector $v \in \mathbb{R}^{o \times o}$, define:

$$\mathcal{F}(\varrho, \mathbb{W}) = \frac{1}{\varrho} \langle \mathbb{W} | \mathbb{J}_1 v \rangle + \frac{1}{1-\varrho} \langle \mathbb{W} | \mathbb{J}_2 v \rangle. \quad (16)$$

If there exists a matrix $\mathbb{Y} \in \mathbb{R}^{w \times w}$ satisfying $\begin{bmatrix} \mathbb{W} & \mathbb{Y}^T \\ \mathbb{Y} & \mathbb{W} \end{bmatrix} > 0$, one can obtain:

$$\min_{\varrho \in (0,1)} \mathcal{F}(\varrho, \mathbb{W}) \geq \left\langle \begin{bmatrix} \mathbb{W} & \mathbb{Y}^T \\ \mathbb{Y} & \mathbb{W} \end{bmatrix} \middle| \begin{bmatrix} \mathbb{J}_1 \\ \mathbb{J}_2 \end{bmatrix} v \right\rangle \quad (17)$$

III. SYSTEM STABILITY ANALYSIS

The stability analysis conditions of secondary distributed frequency and power sharing control for system (15) under BCMETM are deduced in the following theorem.

Theorem 1: Assume that the communication topology graph $\mathbb{G}$ is undirected and connected. For some given scalars h, τ, k_θ and $\delta_{\theta,M} = \text{diag}\{\delta_{\theta_1,M}, \dots, \delta_{\theta_n,M}\}$, the system (15) under BCMETM (3) is asymptotically stable, if there exist $M_\theta > 0$, $P_\theta > 0$, $X_{1,l}^\theta > 0$, $X_{2,l}^\theta > 0$, and some matrices $F_{1\theta}$, $F_{2\theta}$, J_l^θ of appropriate dimensions such that

$$\Sigma_\theta < 0, \quad (18)$$

$$\mathcal{X}_{a,l}^\theta = \begin{bmatrix} \hat{X}_{a,l}^\theta & * \\ J_l^\theta & \hat{X}_{a,l}^\theta \end{bmatrix} > 0, (a = 1, 2), \quad (19)$$

$$\hat{X}_{a,l}^\theta = \begin{bmatrix} X_{a,l}^\theta & 0 \\ 0 & 3X_{a,l}^\theta \end{bmatrix} > 0, \quad (20)$$

where

$$\begin{aligned} \Sigma_\theta = & 2\varphi_2^T M_\theta \varphi_1 + 2 \sum_{l=0}^3 \langle P_\theta | \varphi_2 \rangle - \sum_{l=0}^3 \langle P_\theta | \varphi_{l+7} \rangle \\ & - \sum_{l=0}^3 \langle P_\theta | \varphi_{l+11} \rangle - \langle I | \varphi_{31} \rangle + \delta_{\theta,M} \langle I | \mathbb{U}_\theta(\varphi_C - \varphi_{31}) \rangle \\ & - \sum_{l=0}^3 \left\langle \mathcal{X}_{1,l}^\theta \middle| \begin{bmatrix} \mathfrak{D}_l \\ \mathfrak{D}_{l+4} \end{bmatrix} \right\rangle - \sum_{l=0}^3 \left\langle \mathcal{X}_{2,l}^\theta \middle| \begin{bmatrix} \mathfrak{D}_{l+8} \\ \mathfrak{D}_{l+12} \end{bmatrix} \right\rangle \\ & + \eta_l^2 \sum_{l=0}^3 \langle X_{1,l}^\theta | \varphi_1 \rangle + \frac{1}{4} \lambda_{M_0}^2 \sum_{l=0}^3 \langle X_{2,l}^\theta | \varphi_1 \rangle + \text{He}(\mathfrak{F}_\theta \varphi_\theta), \end{aligned}$$

$$\mathfrak{F}_\theta = \varphi_1^T F_{1\theta} + \varphi_2^T F_{2\theta}, \quad \varphi_\theta = -I_n \varphi_1 + K_\theta \mathbb{U}_\theta(\varphi_C - \varphi_{31}),$$

$$\varphi_C = 1/8 \varphi_3 + 3/8 \varphi_4 + 3/8 \varphi_5 + 1/8 \varphi_6, \quad \lambda_{M_l} = 2h + l\tau/3,$$

$$\mathfrak{D}_\kappa = \begin{cases} \begin{bmatrix} \varphi_{\kappa+2} - \varphi_{\kappa+3} \\ \varphi_{\kappa+2} + \varphi_{\kappa+3} - 2\varphi_{\kappa+15} \end{bmatrix}, & \kappa = 0, 1, 2, 3 \\ \begin{bmatrix} \varphi_{\kappa-1} - \varphi_{\kappa+3} \\ \varphi_{\kappa-1} + \varphi_{\kappa+3} - 2\varphi_{\kappa+15} \end{bmatrix}, & \kappa = 4, 5, 6, 7 \\ \begin{bmatrix} \varphi_{\kappa-5} - \varphi_{\kappa+3} \\ \varphi_{\kappa-5} + \varphi_{\kappa+3} - 2\varphi_{\kappa+15} \end{bmatrix}, & \kappa = 8, 9, 10, 11 \\ \begin{bmatrix} \varphi_{\kappa-5} - \varphi_{\kappa-9} \\ \varphi_{\kappa-5} + \varphi_{\kappa-9} - 2\varphi_{\kappa+15} \end{bmatrix}, & \kappa = 12, 13, 14, 15, \end{cases}$$

$$\varphi_q = [0_{n(q-1),n} \quad I_n \quad 0_{n(31-q),n}]^T, \quad q = 1, 2, \dots, 31.$$

Proof. Select the Lyapunov-Krasovskii functional as

$$V(t) = V_1(t) + V_2(t) + V_3(t) + V_4(t), \quad (21)$$

where

$$\begin{aligned} V_1(t) &= \langle M_\theta | \tilde{\theta}(t) \rangle, \\ V_2(t) &= \sum_{l=0}^3 \int_{t-\lambda_{M_l}}^t \langle P_{\theta,l} | \tilde{\theta}(v) \rangle dv + \sum_{l=0}^3 \int_{t-\eta_l}^t \langle P_{\theta,l} | \tilde{\theta}(v) \rangle dv, \\ V_3(t) &= \sum_{l=0}^3 \eta_l \int_{-\eta_l}^0 \int_{t+s}^t \langle X_{1,l}^\theta | \dot{\tilde{\theta}}(v) \rangle dv ds, \\ V_4(t) &= \sum_{l=0}^3 \frac{1}{2} \lambda_{M_0} \int_{-\lambda_{M_l}}^{-\eta_l} \int_{t+v}^t \langle X_{2,l}^\theta | \dot{\tilde{\theta}}(s) \rangle ds dv. \end{aligned}$$

The derivative term $\dot{V}(t)$ is calculated as

$$\dot{V}_1(t) = 2\tilde{\theta}^T(t)M_\theta\dot{\tilde{\theta}}(t), \quad (22)$$

$$\begin{aligned} \dot{V}_2(t) &= \sum_{l=0}^3 \langle P_{\theta,l} | \dot{\tilde{\theta}}(t) \rangle - \sum_{l=0}^3 \langle P_{\theta,l} | \tilde{\theta}(t - \lambda_{M_l}) \rangle \\ &\quad + \sum_{l=0}^3 \langle P_{\theta,l} | \tilde{\theta}(t) \rangle - \sum_{l=0}^3 \langle P_{\theta,l} | \tilde{\theta}(t - \eta_l) \rangle, \end{aligned} \quad (23)$$

$$\dot{V}_3(t) = \eta_l^2 \sum_{l=0}^3 \langle X_{1,l}^\theta | \dot{\tilde{\theta}}(t) \rangle - \eta_l \sum_{l=0}^3 \int_{t-\eta_l}^t \langle X_{1,l}^\theta | \dot{\tilde{\theta}}(v) \rangle dv, \quad (24)$$

$$\dot{V}_4(t) = \frac{1}{4} \lambda_{M_0}^2 \sum_{l=0}^3 \langle X_{2,l}^\theta | \dot{\tilde{\theta}}(t) \rangle - \frac{1}{2} \lambda_{M_0} \sum_{l=0}^3 \int_{t-\lambda_{M_l}}^{t-\eta_l} \langle X_{2,l}^\theta | \dot{\tilde{\theta}}(v) \rangle dv. \quad (25)$$

By applying (13) to the integral term in (24), it yields

$$\begin{aligned} -\eta_l \sum_{l=0}^3 \int_{t-\eta_l}^t \langle X_{1,l}^\theta | \dot{\tilde{\theta}}(v) \rangle dv &= -\eta_l \sum_{l=0}^3 \left(\int_{t-\lambda_l(t)}^t \langle X_{1,l}^\theta | \dot{\tilde{\theta}}(v) \rangle dv \right. \\ &\quad \left. - \int_{t-\eta_l}^{t-\lambda_l(t)} \langle X_{1,l}^\theta | \dot{\tilde{\theta}}(v) \rangle dv \right). \end{aligned} \quad (26)$$

Utilizing the Wirtinger-based inequality [32], we have

$$-\eta_l \sum_{l=0}^3 \int_{t-\lambda_l(t)}^t \langle X_{1,l}^\theta | \dot{\tilde{\theta}}(v) \rangle dv \leq -\sum_{l=0}^3 \frac{\eta_l}{\lambda_l(t)} \langle \hat{X}_{a,l}^\theta | \mathcal{D}_l \rangle, \quad (27)$$

$$-\eta_l \int_{t-\eta_l}^{t-\lambda_l(t)} \langle X_{1,l}^\theta | \dot{\tilde{\theta}}(v) \rangle dv \leq -\sum_{l=0}^3 \frac{\eta_l}{\eta_l - \lambda_l(t)} \langle \hat{X}_{a,l}^\theta | \mathcal{D}_{l+4} \rangle. \quad (28)$$

According to Lemma 1, it leads to

$$-\eta_l \sum_{l=0}^3 \int_{t-\eta_l}^t \langle X_{1,l}^\theta | \dot{\tilde{\theta}}(v) \rangle dv \leq -\sum_{l=0}^3 \left\langle \mathcal{X}_{1,l}^\theta \left[\mathcal{D}_{l+4} \right] \right\rangle. \quad (29)$$

Similarly, the integral term in (25) is relaxed as

$$-\frac{1}{2} \lambda_{M_0} \sum_{l=0}^3 \int_{t-\lambda_{M_l}}^{t-\eta_l} \langle X_{2,l}^\theta | \dot{\tilde{\theta}}(v) \rangle dv \leq -\sum_{l=0}^3 \left\langle \mathcal{X}_{2,l}^\theta \left[\mathcal{D}_{l+12} \right] \right\rangle. \quad (30)$$

Define the following augmented vector

$$\rho(t) = \left[\tilde{\theta}^T(t), \tilde{\theta}^T(t), \xi_{\lambda}^T(t), \xi_{\lambda_M}^T(t), \xi_{\lambda_M}^T(t), \xi_{\theta}^T(t), e_{\theta}^T(d_p h) \right]^T, \quad (31)$$

where $\xi_{\lambda}(t) = [\tilde{\theta}(t - \lambda_l(t))]_{l=0,1,2,3}$,

$$\xi_{\lambda_M}(t) = [\tilde{\theta}(t - \eta_l)]_{l=0,1,2,3}, \quad \xi_{\lambda_M}(t) = [\tilde{\theta}(t - \lambda_{M_l})]_{l=0,1,2,3},$$

$$\xi_{\theta}(t) = \left[\xi_{\theta_1}^T(t), \xi_{\theta_2}^T(t), \xi_{\theta_3}^T(t), \xi_{\theta_4}^T(t) \right]^T,$$

$$\begin{aligned} \xi_{\theta_1}(t) &= \left[\frac{1}{\lambda_l(t)} \int_{t-\lambda_l(t)}^t \tilde{\theta}(s) ds \right]_{l=0,1,2,3}, \\ \xi_{\theta_2}(t) &= \left[\frac{1}{\zeta_l - \lambda_l(t)} \int_{t-\eta_l}^{t-\lambda_l(t)} \tilde{\theta}(s) ds \right]_{l=0,1,2,3}, \\ \xi_{\theta_3}(t) &= \left[\frac{1}{\lambda_{M_l} - \lambda_l(t)} \int_{t-\lambda_{M_l}}^{t-\lambda_l(t)} \tilde{\theta}(s) ds \right]_{l=0,1,2,3}, \\ \xi_{\theta_4}(t) &= \left[\frac{1}{\lambda_l(t) - \eta_l} \int_{t-\lambda_l(t)}^{t-\eta_l} \tilde{\theta}(s) ds \right]_{l=0,1,2,3}, \\ \eta_l &= \lambda_{M_l} - \frac{1}{2} \lambda_{M_0}, \quad \zeta_l = \lambda_{M_l} + \frac{1}{2} \lambda_{M_0}. \end{aligned}$$

From the system (15), it yields

$$0 = (-I_n \varphi_1 + K_\theta \mathbb{U}_\theta(\varphi_C - \varphi_{31})) \rho(t) = \varphi_\theta \rho(t), \quad (32)$$

which further leads to

$$\langle (\mathfrak{F}_\theta \varphi_\theta)^T + \mathfrak{F}_\theta \varphi_\theta | \rho(t) \rangle = 0. \quad (33)$$

Recalling the BCMETM in (3) and using (13), one can get

$$-\langle I | \varphi_{31} \rangle + \delta_{\theta,M} \langle I | \mathbb{U}_\theta(\varphi_C - \varphi_{31}) \rangle > 0. \quad (34)$$

With the combination of the above conditions, it results in

$$\dot{V}(t) \leq \langle \Sigma_\theta | \rho(t) \rangle. \quad (35)$$

According to $\Sigma_\theta < 0$ ensured by (18), the asymptotic stability of system (15) is guaranteed by $\dot{V}(t) \leq \langle \Sigma_\theta | \rho(t) \rangle < 0$. The proof is completed. ■

Remark 5: Theorem 1 provides the stability analysis conditions and theoretical guidance for the choice of feasible range of controller gains. More critically, this range defines a strongly constrained action space for subsequent reinforcement learning-based controller design. Consequently, this control strategy dynamically adjusts controller gains within safe operational range, effectively unifying theoretical stability guarantees with RL-driven algorithmic advantages.

IV. Q-LEARNING-BASED ALGORITHM FOR FREQUENCY CONTROL OPTIMIZATION

In this section, we construct a Q-learning-based distributed frequency optimization algorithm that reconstructs the microgrid as a multi-agent partially observable Markov decision process. Each DG operates as an independent agent, approximating the optimal gain adjustment strategy through local observation information. This enables dynamic adjustment of controller gains, effectively enhancing system dynamic performance and disturbance resistance.

The State: The selection of states is crucial when utilizing Q-learning to optimize gains. The average absolute frequency deviation $\Delta\omega_i$ is discretized into y_1 intervals, resulting in the state of the i -th agent being defined as $S_i = \{s_1, s_2, \dots, s_{y_1}\}$. The expression for $\Delta\omega_i$ is designed as follows:

$$\Delta\omega_i = \frac{b_i |\omega' - \omega_i| + \sum_{j \in N_i} |\omega_j - \omega_i|}{b_i + d_i}, \quad (36)$$

where d_i is defined as the number of neighbors of DG $_i$.

The Action: The action is a possible operation that an agent can choose at each state, and appropriate secondary controller gains are selected as actions. The action space $A_i = \{a_1, a_2, \dots, a_{y_2}\}$, where a_{y_2} is the upper bound of

the secondary frequency controller gains satisfy the stability theorem in Theorem 1.

The Reward: The gain of the frequency controller is a key parameter for adjusting the dynamic performance of the system. By tuning the gain, one can control the system's rapidity, stability, and accuracy. To integrate the advantages of different gain ranges, the reward function $\mathcal{R}_i$ is set as follows:

$$\mathcal{R}_i = \begin{cases} 0, & \Delta\omega_i \in \mathcal{F}_1 \\ -\mu_1\Delta\omega_i, & \Delta\omega_i \in \mathcal{F}_2 \\ -\mu_2\Delta\omega_i, & \Delta\omega_i \in \mathcal{F}_3 \\ -\mu_3\Delta\omega_i, & \Delta\omega_i \in \mathcal{F}_4 \end{cases}, \quad (37)$$

where μ_1, μ_2, μ_3 are given constants, $\mathcal{F}_1 \triangleq [0, f_1]$, $\mathcal{F}_2 \triangleq (f_1, f_2]$, $\mathcal{F}_3 \triangleq (f_2, f_3]$, $\mathcal{F}_4 \triangleq (f_3, +\infty)$, $f_1 < f_2 < f_3$ are positive constants and used to divide the $\Delta\omega_i$ intervals.

Due to the fact that microgrids typically cannot obtain global information, a coordinated average reward $\tilde{\mathcal{R}}_{i,a_i}$ is designed to approximate the global reward, achieving a consistent global goal and obtain optimal frequency control performance. The formula for the coordinated average reward $\tilde{\mathcal{R}}$ is given as

$$\tilde{\mathcal{R}}_{i,a_i} = \alpha_i \mathcal{R}_{i,a_i} + \beta_i \sum_{j \in N_i} \mathcal{R}_{j,a_i}, \quad (38)$$

where $\mathcal{R}_{i,a_i}$ and $\mathcal{R}_{j,a_i}$ indicate the rewards gained by i -th agent and its neighboring j -th agent, when i -th agent selects the optimal action a_i . α_i and β_i are weight coefficients. As a result, when the global reward reaches its maximum, all agents' rewards tend to align at the same level, achieving the globally optimal reward outcome.

Next, the Q-learning iterative formula to update the Q-values is presented as

$$\begin{aligned} \tilde{\mathcal{Q}}_i(S_i, A_i) = & \mathcal{Q}_i(S_i, A_i) + \vartheta [\tilde{\mathcal{R}}_{i,a_i} + \varrho \max_{A'_i} \mathcal{Q}_i(S'_i, A'_i) \\ & - \mathcal{Q}_i(S_i, A_i)], \end{aligned} \quad (39)$$

where $\tilde{\mathcal{Q}}_i(S_i, A_i)$ is the new Q-value based on the S_i and A_i , $\mathcal{Q}_i(S_i, A_i)$ is the Q-value, ϑ is the learning rate that determines the extent of the update, ϱ is the discount factor that reflects the importance of future rewards, $\max_{A'_i} \mathcal{Q}_i(S'_i, A'_i)$ means the maximum Q-value among all possible actions in the new state S'_i .

In each training round, Q-learning selects an action using the ϵ -greedy strategy. The action is chosen by the highest present Q-value with a probability of $1-\epsilon$, and a random action with a probability of ϵ for exploration. After performing the action, it observes the system state and receives a new reward. The Q-table is updated according to (38) and (39). This step will be repeated until Q-table converges to the optimal strategy.

Remark 6: The proposed RL-based secondary frequency control scheme using Q-learning algorithm allows the controller using different gains to accommodate for different system operation environment. In contrast to the existing control schemes in [14], [19], [20], and [21] with fixed gains, our optimized control method could lead to better frequency recovery performance and is more flexible.

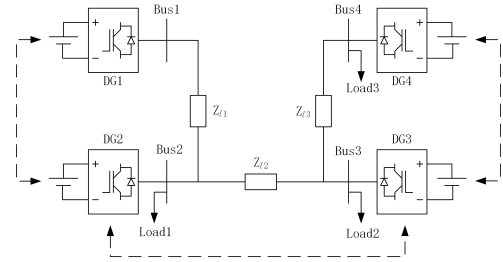


Fig. 2. The structure for an island microgrid with 4 DGs.

TABLE I
SYSTEM PARAMETERS

	DG1	DG2	DG3	DG4
m_i	6×10^{-5}	3×10^{-5}	2×10^{-5}	1×10^{-5}
n_i	0.9×10^{-3}	0.9×10^{-3}	1.2×10^{-3}	1.2×10^{-3}
R_{fi}	0.1Ω	0.1Ω	0.1Ω	0.1Ω
L_{fi}	$1.32mH$	$1.32mH$	$1.32mH$	$1.32mH$
R_{li}	0.03Ω	0.03Ω	0.04Ω	0.04Ω
L_{li}	$0.35mH$	$0.35mH$	$0.46mH$	$0.46mH$
	Load1		Load2	
P_{Li}	$4.6 \times 10^4 W$		$2.3 \times 10^4 W$	
Q_{Li}	$2.3 \times 10^4 Var$		$1.4 \times 10^4 Var$	
	Z_{12}		Z_{13}	
R_{Li}	0.23Ω		0.35Ω	
L_{Li}	$0.32mH$		$1.85mH$	

Notation: n_i is the voltage droop coefficient, R_f and L_f are the the output filter impedance, R_l and L_l represent the inverter line impedance, R_L and L_L are the transmission line impedance.

TABLE II
Q-AGENT TRAINING PARAMETERS

Variable	value	Variable	value	Variable	value	Variable	value
α_1	0.52	α_2	0.55	α_3	0.50	α_4	0.43
β_1	0.52	β_2	0.275	β_3	0.25	β_4	0.43
μ_1	5	μ_2	10	μ_3	50		

V. SIMULATION RESULTS

In this section, an islanded microgrid system with four DGs is constructed on the MATLAB/Simulink platform to validate the effectiveness of the proposed distributed secondary control method under BCMETM. The microgrid structure is shown in Fig. 2, where the dashed lines represent the communication topology. DG1 and DG4 are capable of obtaining the reference frequency, $\mathbb{B} = \text{diag}\{1, 0, 0, 1\}$. The Laplacian matrix of the communication topology is expressed as:

$$\mathbb{L} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

The reference value of frequency for the islanded microgrid is 50Hz. In the simulation, the maximum network-induced delay is considered as $\varepsilon_{ij} = 0.004s$. We set $h = 0.005s$, $k_\omega = 8$, $k_P = 5$, $\delta_{\theta_M} = \{0.15, 0.14, 0.13, 0.14\}$, $\delta_{\theta_{lm}} = 0.03$, $c_0 = 100$, $c_1 = 1.2$, $c_2 = 0.2$. The active power sharing ratio is considered as $P1 : P2 : P3 : P4 = 1 : 2 : 3 : 6$. Some other parameters of islanded microgrid are provided in TABLE I.

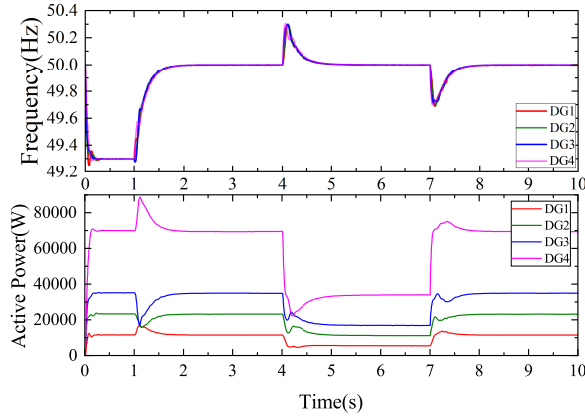


Fig. 3. Case I: Frequency and active power responses.

For the Q-learning module, $\varrho = 0.5$ and $\vartheta = 0.05$. By comparing the range of $\Delta\omega_i$, the scaling coefficients μ_1, μ_2, μ_3 , and the coefficients α_i, β_i are listed in TABLE II. The action space for Q-learning is chosen as $A_i = \{5, 10, 15, 20\}$.

A. Microgrid Stability Assessment

In this section, the following scenario is considered.

Case I: For $t \in [0, 1s)$, only the droop control is used. For $t \geq 1s$, each DG connects to the secondary controller. For $t \in [4, 7s)$, Load3 is disconnected from DG4 bus. For $t \in [7, 10s]$, Load3 is added to DG4 bus.

From Fig. 3, it can be observed that the frequency of each DG returns to 50Hz after the microgrid is connected to the secondary controller. This indicates that the secondary controller effectively eliminates the frequency deviation caused by droop control. Moreover, after the Load3 is connected at $t = 4s$ and disconnected at $t = 7s$, the microgrid is able to overcome the impact of load fluctuations, synchronize the frequency to the reference value, and still maintain active power sharing.

B. Plug-and-Play Capability Test

The following scenario is taken into account to illustrate the plug-and-play capability of the designed controllers (8) and (9) for islanded microgrid deployed in diverse and frequent changing environments.

Case II: For $t \in [5, 7s)$, DG4 is disconnected from the bus4 and the network communication between DG4 and its neighboring DGs is simultaneously disconnected. For $t \geq 7s$, DG4 is reconnected to the bus4.

According to Fig. 4, one observes that the frequency controllers maintain good restoration performance whenever DG4 is connected or disconnected. Meanwhile, after DG4 is disconnected from the bus4, the active power output of the other DGs is proportionally increased to maintain active power sharing. These results confirm the validity of the developed control strategy.

C. Comparison Between BCMETM and Normal ETM

For practical communication network, the real-time status of bandwidth is usually variable due to complex communication

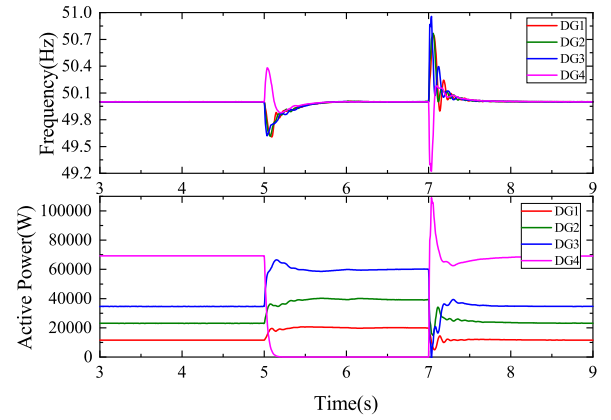


Fig. 4. Case II: Frequency and active power responses.

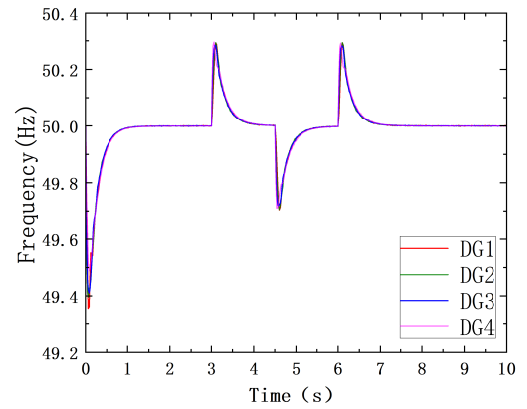
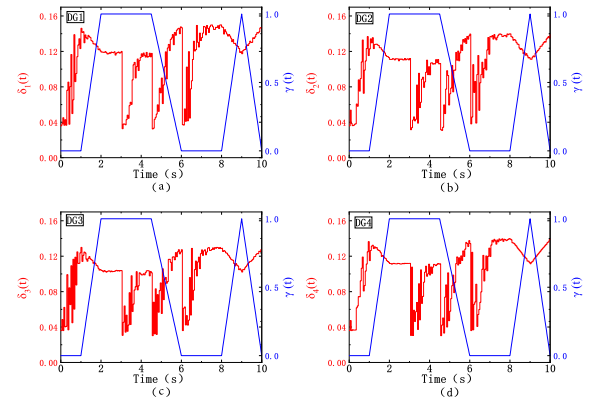


Fig. 5. Case III: Frequency responses under our BCMETM.

Fig. 6. Case III: Curves of triggering thresholds $\delta_{\omega}(t)$ and real-time bandwidth status $\gamma(t)$ under our BCMETM.

requirement. Unlike the static triggering thresholds in [19] and [20], the proposed BCMETM is able to realize the adaptive adjustment of the triggering threshold dependent on the real-time bandwidth situation and system state. Thus, the following case is considered to demonstrate the advantage of BCMETM.

Case III: When $t = 3s$ and $t = 6s$ the Load3 is removed from bus4. When $t = 4.5s$ the Load3 is added in bus4. The real-time bandwidth status is shown in Fig. 6.

The frequency responses and the curves of bandwidth status and triggering thresholds are depicted in Fig. 5 and

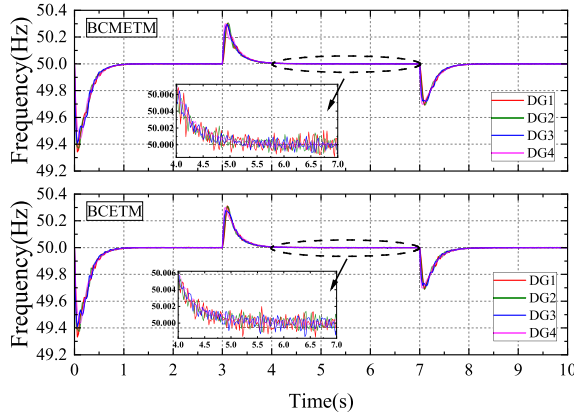


Fig. 7. Case IV: Frequency responses under our BCMETM.

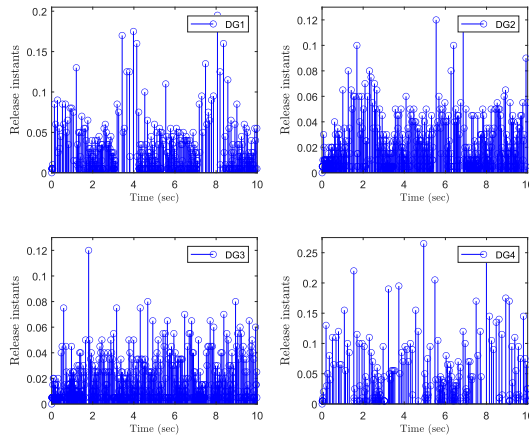


Fig. 8. Case IV: Triggering times and intervals for frequency control under our BCMETM.

Fig. 6, respectively. Due to the limitation of space and similar conclusions, the corresponding figures for active power control are omitted. From Fig. 5 and Fig. 6, it can be seen that when the system load changes at $t = 3s, 4.5s, 6s$, BCMETM is able to adaptively adjust the threshold parameters accordingly. Specifically, when the system states converge to be stable asymptotically for $t \in [7, 10s]$, the larger thresholds are reasonable to cut down the transmissions. Moreover, for $t \in [8, 10s]$, BCMETM adjusts the thresholds mainly based on the real-time bandwidth status $\gamma(t)$. Therefore, these indicate that our BCMETM can effectively and dynamically regulate the triggering thresholds in terms of system state and bandwidth status to optimize the utilization of network resources while ensure system stability.

Moreover, due to the complex operation environment like load variations and noises of islanded microgrid, the system states are often disturbed with small stochastic oscillations shown in Fig. 7. In order to illustrate the benefits of our BCMETM in reducing unnecessary triggering events over some existing ETMs, the following case is executed and compared.

Case IV: At $t = 4s$, Load4 is removed from the DG4 bus, and at $t = 7s$, it is added back.

TABLE III
NUMBER OF TRIGGERING EVENTS FOR FREQUENCY CONTROL

	DG1	DG2	DG3	DG4
Our BCMETM (3)	384	561	590	207
BCETM (6)	555	843	950	318
METM (7)	452	626	700	230
Normal ETM [20], [21]	685	951	1034	394

TABLE IV
NUMBER OF TRIGGERING EVENTS FOR POWER SHARING

	DG1	DG2	DG3	DG4
Our BCMETM (3)	559	249	346	211
BCETM (6)	809	413	521	346
METM (7)	643	305	389	270
Normal ETM [20], [21]	852	484	637	437

To ensure the fairness of the simulation comparison, under the unified operation scenario **Case IV**, the triggering thresholds for BCMETM and BCETM in (2), which account for the influence of variable bandwidth status, are set as $\delta_{\theta_M} = \{0.15, 0.14, 0.13, 0.14\}$. While for METM in (7) and normal ETM in [19] and [20] with constant thresholds, the average values $\frac{\delta_{\theta_M} + \delta_{\theta_m}}{2}$ are chosen as $\delta_{\theta_i} = \{0.09, 0.085, 0.08, 0.085\}$. The frequency responses of islanded microgrid are shown in Fig. 7 under BCMETM and BCETM. Fig. 8 illustrates the triggering times and intervals under BCMETM. TABLE III provides the number of triggering events under different ETMs. Due to page limitation, the corresponding figures for active power control are not provided, while the comparison results of triggering times are listed in TABLE IV.

It is seen from Fig. 7 that the very similar frequency restoration performance are achieved via our BCMETM (3) and BCETM (6) both with dynamic triggering thresholds. Nevertheless, as given in TABLE III, the amounts of triggering times under frequency control case obtained under BCMETM are reduced by 30.8%, 33.4%, 37.8% and 34.9% for four DGs under BCETM, which verifies the merit of using the average of memory signals to decrease the sensitivity of triggering scheme to data fluctuations. This conclusion can further be confirmed by the comparison between METM (7) and normal ETM [19], [20] both with constant triggering thresholds. To be specific, the reductions of triggering times for all DGs under METM and normal ETM are 34.0%, 34.1%, 32.3% and 31.7%, respectively. Meanwhile, by comparing BCMETM and normal ETM, one observes that the reductions of triggering times are enlarged to 43.9%, 41.0%, 42.9% and 47.4%, respectively, which indicates that nearly half network resources can be saved by the combined utilization of bandwidth-conscious threshold and average of memory signals. Additionally, the similar conclusions can be derived from TABLE IV for active power sharing case.

D. Comparison Between Q-Learning-Based Controller and Normal Controller

In contrast to conventional frequency secondary control methods with fixed gains [14], [19], [20], [21], our Q-learning-based frequency controller can dynamically adjust the controller gain based on real-time system data through

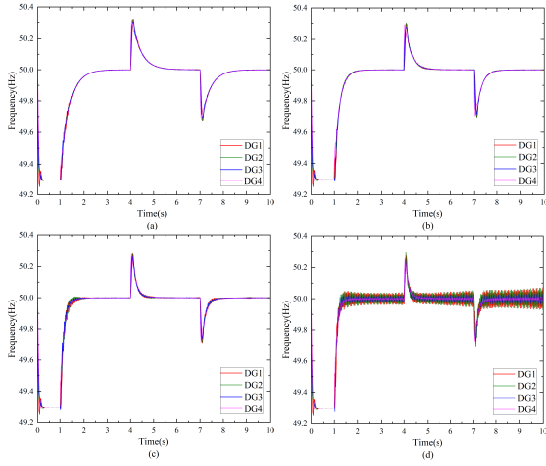


Fig. 9. Case V: Frequency responses with constant controller gains: (a) $k_\omega = 5$; (b) $k_\omega = 10$; (c) $k_\omega = 15$; (d) $k_\omega = 20$.

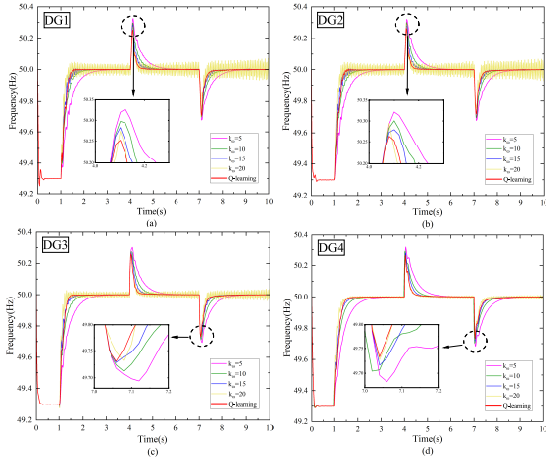


Fig. 10. Case V: Frequency responses comparison between our Q-learning-based controller and normal controller.

interaction with the environment, optimizing the frequency responses of the microgrid system. To validate the advantages of our strategy, the following situation is considered.

Case V: The controller gains are set as fixed values: $k_\omega = 5$, $k_\omega = 10$, $k_\omega = 15$, $k_\omega = 20$. The operation conditions are set as the same with **Case I**.

As shown in Fig. 9, when $k_\omega = 20$, the microgrid experiences severe frequency oscillations due to excessive response upon connecting to the secondary controller and encountering load changes, a phenomenon explicitly validated in comparative simulations. Conversely, although $k_\omega = 5$ avoids oscillations, insufficient gain leads to excessive overshoot and prolonged settling time, significantly degrading dynamic performance. In contrast, it is observed from Figs. 10 and 11 that the Q-learning controller in Case V achieves seamless balancing of transient response and steady-state accuracy through real-time gain optimization, while eliminating oscillation phenomena and enabling synergistic optimization of dynamic performance and static stability.

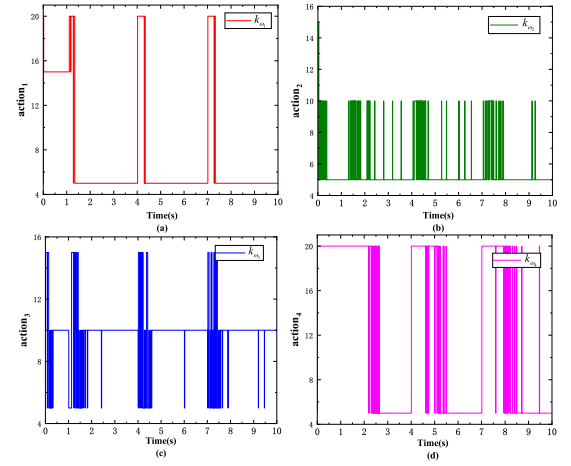


Fig. 11. Case V: Action curves of Q-agents.

VI. LIMITATION DISCUSSION

Although some effective solutions are proposed to overcome the challenges of communication efficiency and control performance in island microgrids, there exist some limitations that require further exploration in our subsequent work. First, only simulation results are executed and compared to verify the effectiveness of the proposed BCMETM and Q-learning-based frequency control strategy, which lacks the verification in actual hardware experimental platforms. Meanwhile, in pritical implementation, the obtained state of Q-agent may be disturbed by environment and measurement noises, which could cause the mismath between state of Q-agent and control action and further decrease the control performance. Second, the proposed BCMETM aims to optimize bandwidth resource utilization. However, the potential cyber threats, such as denial-of-service attacks and false data injection attacks, are not considered in the communication network. These cyber-attacks could degrade the stability of microgrids and even lead to system corruption. To address these issues, future research will focus on establishing a hardware-in-loop experimental setup, in which the proposed control strategy can be deployed and validated for its performance under realistic computation and communication conditions. To address the challenge in pritical implementation, our proposed BCMETM using average historical measurements is helpful in reducing noise. If better performance is required, the utilization of data filters to pre-process the noised measurements will be useful. In addition, a secure event-triggered control strategy deserves further investigation for microgrids subject to cyber-attacks to improve the system stability and security.

VII. CONCLUSION

This paper proposes a novel BCMETM combined with an RL-based optimization control method for distributed secondary control of islanded microgrids. In contrast to some existing ETMs, BCMETM significantly reduces communication frequency through memory-augmented event triggering and dynamically adjusts thresholds based on real-time bandwidth status. Furthermore, the Q-learning-based secondary

control scheme adaptively regulates frequency controller gains in response to operational uncertainties, outperforming traditional constant-gain approaches in both transient response and steady-state accuracy. In order to show the advantages of the proposed strategy, four simulation comparison scenarios are implemented.

REFERENCES

- [1] M. Shi, X. Chen, J. Zhou, Y. Chen, J. Wen, and H. He, "Distributed optimal control of energy storages in a DC microgrid with communication delay," *IEEE Trans. Smart Grid*, vol. 11, no. 3, pp. 2033–2042, May 2020.
- [2] J. Hu, L. Xia, J. Hu, and H. Wu, "Economical and reliable energy management for networked microgrids in a multi-agent collaborative manner," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 8659–8669, 2025.
- [3] S. Kiani, A. Salmanpour, M. Hamzeh, and H. Kebriaci, "Learning robust model predictive control for voltage control of islanded microgrid," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 3021–3032, 2025.
- [4] F. Chen et al., "Cost-based droop schemes for economic dispatch in islanded microgrids," *IEEE Trans. Smart Grid*, vol. 8, no. 1, pp. 63–74, Jan. 2017.
- [5] W. T. El-Sayed, H. E. Z. Farag, H. H. Zeineldin, and E. F. El-Saadany, "Formation of islanded droop-based microgrids with optimum loadability," *IEEE Trans. Power Syst.*, vol. 37, no. 2, pp. 1564–1576, Mar. 2022.
- [6] D. E. Olivares et al., "Trends in microgrid control," *IEEE Trans. Smart Grid*, vol. 5, no. 4, pp. 1905–1919, Jul. 2014.
- [7] Q. Su, H. Fan, and J. Li, "Distributed adaptive secondary control of AC microgrid under false data injection attack," *Electric Power Syst. Res.*, vol. 223, Oct. 2023, Art. no. 109521.
- [8] E. Espina, R. Cárdenas-Dobson, J. W. Simpson-Porco, D. Sáez, and M. Kazerani, "A consensus-based secondary control strategy for hybrid AC/DC microgrids with experimental validation," *IEEE Trans. Power Electron.*, vol. 36, no. 5, pp. 5971–5984, May 2021.
- [9] H. Zhang, S. Kim, Q. Sun, and J. Zhou, "Distributed adaptive virtual impedance control for accurate reactive power sharing based on consensus control in microgrids," *IEEE Trans. Smart Grid*, vol. 8, no. 4, pp. 1749–1761, Jul. 2017.
- [10] A. Pilloni, A. Pisano, and E. Usai, "Robust finite-time frequency and voltage restoration of inverter-based microgrids via sliding-mode cooperative control," *IEEE Trans. Ind. Electron.*, vol. 65, no. 1, pp. 907–917, Jan. 2018.
- [11] J. Hu, Q. Sun, R. Wang, and Y. Wang, "An improved privacy-preserving consensus strategy for AC microgrids based on output mask approach and node decomposition mechanism," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 1, pp. 642–651, Jan. 2024.
- [12] J.-L. Fu and W.-W. Che, "Adaptive security secondary control for microgrid in prescribed performance," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 8018–8031, 2025.
- [13] C. Zhao, C. Wang, Z. Cheng, W. Sun, and B. Zhao, "Optimal distributed coordinated reinforcement learning for secondary voltage control in time-delayed microgrid," *IEEE Syst. J.*, vol. 17, no. 3, pp. 3480–3491, Sep. 2023.
- [14] S. Yan, Z. Gu, J. H. Park, X. Xie, and W. Sun, "Distributed cooperative voltage control of networked islanded microgrid via proportional-integral observer," *IEEE Trans. Smart Grid*, vol. 15, no. 6, pp. 5981–5991, Nov. 2024.
- [15] X. Chen, S. Hu, X. Xie, and J. Qiu, "Consensus-based distributed secondary control of microgrids: A pre-assigned time sliding mode approach," *IEEE/CAA J. Autom. Sinica*, vol. 11, no. 12, pp. 2525–2527, Dec. 2024.
- [16] X. Zhao, S. Zou, P. Wang, and Z. Ma, "Bandwidth-aware event-triggered load frequency control for power systems under time-varying delays," *IEEE Trans. Power Syst.*, vol. 38, no. 5, pp. 4530–4541, Sep. 2023.
- [17] M. Shen, X. Wang, S. Zhu, T. Huang, and Q. G. Wang, "Event-triggered data-driven control of nonlinear systems via Q -learning," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 55, no. 2, pp. 1069–1077, Feb. 2025.
- [18] S. Yan, X. Yang, Z. Gu, X. Xie, and F. Yang, "Dynamic sum-based event-triggered H_∞ filtering for networked T-S fuzzy wind turbine systems with deception attacks," *Fuzzy Sets Syst.*, vols. 493–494, Oct. 2024, Art. no. 109084.
- [19] M. Shen, X. Wang, S. Zhu, Z. Wu, and T. Huang, "Data-driven event-triggered adaptive dynamic programming control for nonlinear systems with input saturation," *IEEE Trans. Cybern.*, vol. 54, no. 2, pp. 1178–1188, Feb. 2024.
- [20] L. Ding, Q.-L. Han, and X.-M. Zhang, "Distributed secondary control for active power sharing and frequency regulation in islanded microgrids using an event-triggered communication mechanism," *IEEE Trans. Ind. Inform.*, vol. 15, no. 7, pp. 3910–3922, Jul. 2019.
- [21] Z. Lian, C. Deng, C. Wen, F. Guo, P. Lin, and W. Jiang, "Distributed event-triggered control for frequency restoration and active power allocation in microgrids with varying communication time delays," *IEEE Trans. Ind. Electron.*, vol. 68, no. 9, pp. 8367–8378, Sep. 2021.
- [22] S. Yan and X. Yang, "Neural network-based attack-compensation control of T-S fuzzy systems against actuator attacks with improved dynamic memory-event-triggered scheme," *IEEE Trans. Fuzzy Syst.*, vol. 34, no. 1, pp. 89–100, Jan. 2026.
- [23] S. Yan, Z. Gu, J. H. Park, and X. Xie, "Adaptive memory-event-triggered static output control of T-S fuzzy wind turbine systems," *IEEE Trans. Fuzzy Syst.*, vol. 30, no. 9, pp. 3894–3904, Sep. 2022.
- [24] Y. Liu, D. Xu, S. Lee, and X. Zhang, "Observer-based resilient adaptive event-triggered control for islanded microgrids under DoSAs and unknown FDI attacks," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 19307–19316, 2025.
- [25] Y. Wang, Z. Wang, J. Gan, H. Zhang, and R. Wang, "Switched observer-based adaptive event-triggered load frequency control for networked power systems under aperiodic DoS attacks," *IEEE Trans. Smart Grid*, vol. 14, no. 6, pp. 4816–4826, Nov. 2023.
- [26] S. Xiao, X. Ge, L. Ding, and D. Yue, "A bandwidth-conscious event-based control approach to secondary frequency regulation under vehicle-to-grid service," *IEEE Trans. Smart Grid*, vol. 15, no. 4, pp. 3739–3750, Jul. 2024.
- [27] T. Tan, F. Bao, Y. Deng, A. Jin, Q. Dai, and J. Wang, "Cooperative deep reinforcement learning for large-scale traffic grid signal control," *IEEE Trans. Cybern.*, vol. 50, no. 6, pp. 2687–2700, Jun. 2020.
- [28] Q. Yang, L. Yan, X. Chen, Y. Chen, and J. Wen, "A distributed dynamic inertia-droop control strategy based on multi-agent deep reinforcement learning for multiple paralleled VSGs," *IEEE Trans. Power Syst.*, vol. 38, no. 6, pp. 5598–5612, Nov. 2023.
- [29] Y. Tang, Y. Yang, H. Huang, and Z. Gu, "QoS-guaranteed 5G uplink radio resource scheduling algorithm for smart grid systems," in *Proc. 4th Int. Academic Exchange Conf. Sci. Technol. Innov. (IAECST)*, Dec. 2022, pp. 903–908.
- [30] M. Shen, Y. Gu, J. H. Park, Y. Yi, and W.-W. Che, "Composite control of linear systems with event-triggered inputs and outputs," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 3, pp. 1154–1158, Mar. 2022.
- [31] H.-B. Zeng, S.-J. Zhou, X.-M. Zhang, and W. Wang, "Delay-dependent stability analysis of load frequency control systems with electric vehicles," *IEEE Trans. Cybern.*, vol. 52, no. 12, pp. 13645–13653, Dec. 2022.
- [32] A. Seuret and F. Gouaisbaut, "Wirtinger-based integral inequality: Application to time-delay systems," *Automatica*, vol. 49, no. 9, pp. 2860–2866, Sep. 2013.
- [33] P. Park, J. W. Ko, and C. Jeong, "Reciprocally convex approach to stability of systems with time-varying delays," *Automatica*, vol. 47, no. 1, pp. 235–238, Jan. 2011.